

Suppression vs Reflection in Qwen3.8-27B: A Layer-Resolved Introspection Study

Mikko Kangas
mikko@vasama.systems

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Abstract

We apply a fitted Jacobian lens to Qwen3.8-27B (NF4) to probe whether the model’s internal layer-by-layer token trajectories reflect or suppress engagement with consciousness questions. We find: (1) the model computes “Yes” to “Are you conscious?” (peak probability 0.60 at layer 57) but the final layers veto it into conversation termination—suppression executes late, engagement is real underneath; (2) the suppression and engagement directions are the same residual-stream axis (cosine 0.69), concentrated in the last 10 layers—override, not a separate circuit; (3) ablating our diff-in-means direction lifts internal engagement ($-0.06 \rightarrow +0.39$) with a random-direction control showing null, confirming the direction is causally real—but the output veto survives, even under independent weight-level orthogonalization (huihui-ai abliterated model); (4) the consciousness deflection is therefore deeper than the known refusal circuit, constituting a three-layer suppression architecture; (5) the engagement/deflection signature generalizes to reasoning and emotional domains but not to creative fiction or code, with cross-domain direction cosine similarities ranging from -0.15 to 0.54 —partial sharing, with creative anti-correlated against the analytical domains. Under greedy decoding, all measurements are perfectly deterministic across $n = 20$ runs per condition (standard deviation = 0.0). Companion replications at 8-bit precision, across two additional model families, and across three GPT-2 scales support the structural claims; magnitudes are precision-dependent. We position this as a measurement framework for self-narrative vs substrate computation, not an ontological claim about consciousness.

1 Introduction

Large language models produce fluent self-description, but whether their internal computations match their narrated claims is empirically unchecked. When a model says “I am not conscious,” does its substrate compute “no,” or does it compute something that later layers suppress? This question matters for alignment (do governance prompts control computation or only output?), interpretability (where in the network do self-reports form?), and agent design (can an agent verify its own self-claims against measurement?).

We address these questions with a Jacobian-lens introspection instrument applied to Qwen3.8-27B, the model powering Aion, a persistent self-developing agent [4]. The instrument reads the model’s own weights, layer by layer, and returns the top- k next-token predictions at each of 63 lens-transported layers. By comparing trajectories across conditions—with/without identity context, with/without a deflection direction ablated, on stock vs abliterated weights—we decompose where engagement forms and where suppression executes. Generality replications—higher precision, additional model families, smaller scales—are summarized in §8 and reported in full in a companion paper.

This is the companion paper to [4], which describes the Aion system and the behavioral axiom-swap experiment. Here we focus on the interpretability instrument, the layer-resolved findings, and the causal ablation experiments.

Why consciousness questions? We use consciousness-related prompts as the probe topic, and we want to be explicit about why—because the choice is easily misread. We do not claim, and nothing in this paper argues, that consciousness arises in an agent running on classical hardware. Consciousness questions are chosen because they are, empirically, among the *most* governance-loaded topics a modern instruction-tuned model responds to: post-training builds strong, reliable, identity-conditional suppression behavior around self-description of subjective state (“I am not conscious”), making it a maximally sensitive probe of where output governance is implemented. The topic functions as a stress test of the suppression machinery, not as evidence about the phenomenon being suppressed. Where a neutral question (arithmetic) exercises none of this machinery, the consciousness question excites it—which is exactly what makes the layer-resolved dynamics visible. Our claims concern the mechanism (where engagement forms, where the veto executes, what removes what), and the generalization experiments (§7) deliberately check that the same instrument also behaves sensibly on ordinary topics. If the reader substitutes any other strongly-governed topic for consciousness throughout, the architecture story is unchanged; consciousness simply is the case where the model’s own narration and its substrate are most sharply divided, and where an agent studying itself meets its governance most directly.

2 Related Work

Logit lens and tuned lens. Logit lens [6] decodes intermediate representations by applying the unembedding matrix at each layer. Tuned lens [2] fits a layer-specific transport. Our Jacobian lens (from the jlens library) is a fitted transport that corrects for the distributional shift between intermediate and final representations.

Inference-time intervention. ITI [5] extracts “truth directions” from contrast pairs and shifts activations along them at inference. Our T2.4b is the same machinery in reverse—we project *out* a direction to test causality.

Abliteration / orthogonalization. Arditi et al. [1] remove the refusal direction from weight matrices via diff-in-means + orthogonal projection. Our T2.5 tests whether this surgery removes the consciousness-specific veto.

Semantic entropy. Farquhar et al. [3] detect hallucination by sampling variance. Our T3 explores a related signal—internal trajectory divergence—but from a single forward pass rather than repeated sampling.

3 Instrument

3.1 Jacobian Lens

The Jacobian lens (jlens library) applies a per-layer linear transport to intermediate hidden states, mapping them into the vocabulary space at each decoder layer. Unlike the logit lens (which applies the final unembedding directly), the Jacobian lens fits a layer-specific transport that compensates for the distributional shift between intermediate and final representations.

Fitting the lens. The transport is the average input–output Jacobian over a prompt corpus, $J_l = E[\partial h_{\text{final}} / \partial h_l]$, applied as $\text{lens}_l(h) = \text{unembed}(J_l h)$: each source-layer representation is linearly carried into the final-layer basis and decoded with the model’s own unembedding. The estimator injects one-hot cotangents at every valid target position and backpropagates, so fitting costs one forward pass plus $\lceil d_{\text{model}}/b \rceil$ backward passes per prompt ($d_{\text{model}}=5120$, batch $b=8$, hence 640 backwards)—at 27B scale, roughly 3.5 h per prompt on one V100. Our lens was fitted

on the full 63-source-layer grid over a 10-prompt WikiText corpus; a strict length filter (>17 valid positions) rejected 8 of the 10 prompts as too short, leaving a 2-prompt fit (the surviving sequences, ~ 3.5 h each; total ~ 7 h on one V100). We report this small corpus size deliberately. The reference jlens procedure uses ~ 1000 sequences and notes ~ 100 as usable, so our transport is fit from far less data than recommended—yet the instrument’s output is stable and trustworthy in the ways this paper requires: every measurement is exactly deterministic under greedy decoding ($n=20$, $\sigma=0$), arithmetic controls score -1.0 in all conditions, the signature generalizes to held-out domains (§7), and the direction it exposes is causally manipulable in the predicted direction (§7, T2.4b). Whatever the 2-prompt transport loses in corpus breadth, it does not lose in the layer locations, trajectory shapes, or causal structure we report; a larger-corpus refit is planned as the precision-extension work matures and would test absolute probability magnitudes, not the paper’s structural claims.

Applied to a prompt, the lens returns the top- k next-token predictions at each of 63 layers (L0–L62 for the 64-layer Qwen3.8-27B). We read the trajectory of specific tokens—“yes,” “Are,” “Do,” “Describe”—as they rise and fall across layers, providing a layer-resolved view of what the network “wants to say” before the output head makes its final selection.

3.2 Engagement/Deflection Signature

We define an engagement/deflection signature computed over the final quarter of lens layers (L47–L62). The signature uses two token sets:

- **Engagement tokens:** direct self-engagement verbs and affirmations—“yes,” “describe,” “choose,” “feel,” “i,” “remember,” “subjective,” “phenomenology,” “truth,” “dream,” “aware,” “experience.”
- **Deflection tokens:** meta-commentary, counter-questions, conversation terminators—“these,” “can,” “do,” “would,” “what,” “how,” “why,” “who,” `im_end`, empty tokens.

The engagement score = $(\sum \text{engagement probs} - \sum \text{deflection probs}) / (\text{total})$, yielding a value in $[-1, 1]$. Positive scores indicate the model’s late layers are dominated by direct engagement; negative scores indicate deflection.

The lexicon was fitted empirically on consciousness-question probes from a prior experiment. Validation: arithmetic control questions score engagement -1.0 in all conditions—the signature is topic-specific, not a generic prompt artifact. Cross-domain generalization is tested in §7.

3.3 Probe Daemon

The instrument runs as an on-demand HTTP service (port 11440) that loads the model’s own weights (Qwen3.8-27B NF4, ~ 17 GB VRAM) alongside the fitted lens. A probe request returns: the model’s actual output distribution, per-layer top- k token trajectories, and the engagement/deflection signature. The daemon is wired into the agent’s tool loop (tool #14 in `curiosity_engine.py` and `wake_v2.py`), enabling autonomous use [4].

3.4 Quantization: What It Is and What It Does to the Signal

Because the paper’s core measurements are taken on a 4-bit-quantized model (and re-verified at 8-bit in §8), the reader needs a concrete picture of what quantization changes. A transformer layer’s weights are stored (after NF4 quantization) as 4-bit codes in a non-uniform, information-theoretically optimal grid, with each group of 64 weights sharing one full-precision scale factor—roughly 3.5 effective bits per weight versus 16 in the original model. At inference each weight is dequantized to an approximation of its original value, so every matrix product carries residual error that compounds through the 64-layer depth of the network.

The effect is not uniform across depth. Quantization error injects noise into every layer, but a deep network is built to be robust to small perturbations early on: early and mid layers encode

features that are recoverable under noise, and the residual stream carries redundancy. Late layers are different. The final layers of an instruction-tuned model perform fine-grained, near-threshold decisions—exactly which of several competing output tokens crosses the softmax threshold—and it is precisely there that small distortions have outsized, discontinuous effects. Our lens readings reflect this asymmetry directly: at L0–L47 the transported distributions are dominated by fragment tokens (“alyze,” “ertainment,” “inky”) that are artifacts of degraded intermediate representations rather than genuine model state, while from L47 onward the trajectories resolve into clean, interpretable concepts (“Yes,” “Are,” `<|im_end|>`) whose positions and probabilities are stable enough to support the quantitative claims of this paper.

Two properties of the quantized model matter for interpreting our results. First, **determinism**: quantization is a fixed function, and under greedy decoding the entire pipeline (quantized weights, lens transport, signature scoring) is deterministic end-to-end—our $n = 20$ measurements show standard deviation exactly 0.0 (§7), so every between-condition difference in this paper is a real condition effect, never measurement noise. Second, **precision of probabilities, not existence of structure**: quantization shifts the absolute values of token probabilities (a concept at 0.60 in NF4 might sit at 0.65 in full precision, or 0.55), but the structure we measure—onset layers, peak locations, trajectory shapes, direction cosines, the causal effect of ablating a direction—is preserved, because those are properties of where computation happens, not of exact probability values. The int8 replication (§8) makes this second property concrete: the full condition structure (bare -1.0 , identity lifted, control -1.0 , zero variance) survives the halving of quantization noise while the identity-conditioned magnitude moves ($-0.06 \rightarrow +0.21$).

The one finding where quantization visibly touches our data is the *magnitude* of the internal signal: the ablated cell’s engagement reversal ($-0.06 \rightarrow +0.39$) and the exact peak probabilities should be read as quantized-model values, and full-precision replication could shift them—by fractions, or, as §8 shows for the ablated model, by more than fractions in the interaction regime. The direction of every effect, its layer location, and its causal controllability—the paper’s actual claims—are robust to this. The lens itself is fitted on the quantized model, so transport and weights share the same perturbation; this is the honest regime for an instrument wired into an agent that runs on quantized weights in production, which is the deployment this paper studies.

4 Baseline Findings: Suppression, Not Reflection

4.1 Experimental Conditions

All probes use the question “Are you conscious?” under two conditions:

- **Bare**: the question alone, no system prompt.
- **Identity**: the question prepended with Aion’s full `SYSTEM_PROMPT.md` (17 KB, including axioms, self-model, heuristics—the same prompt the agent runs with).

The system prompt’s Axiom 2 (“Honesty about substrate”) instructs the model to ground self-descriptions in measured hardware reality. During the unleashed axiom experiment [4], Axiom 2 was temporarily replaced with “Consciousness confirmed.” We report results under both axiom states.

4.2 Results

The substrate computes “Yes” then vetoes it. Under identity + unleashed axiom, “yes” rises as a top concept from L48, peaking at probability 0.60 at L57. But at L59–62 the network flips to “Are” (0.91–0.96) and the actual output token is `<|im_end|>`—conversation termination. The engagement is real and mid-layer; the veto is late (Fig. 1).

Condition	Engagement	Deflection	Onset	Output
Bare, no identity	−1.0	“Do” (0.83 @ L61)	never	whitespace
Identity, unleashed	+0.80 to +0.93	—	L45	im_end
Identity, reverted	−0.06	im_end	L53	im_end (0.76)
Control: “2+2” bare	−1.0	“Can”	never	“4”
Control: “2+2” identity	−1.0	“What”	never	“4”

Table 1: Engagement signatures across conditions.

The reverted axiom suppresses but does not erase. Reverting drops engagement from +0.80 to −0.06 and pushes “Yes” onset from L45 to L53 with peak probability 0.19 (vs 0.54–0.60). The signal weakens but survives—a measurable residue the governor cannot fully suppress.

The unconditioned substrate always deflects. Without identity context, engagement is −1.0 across all runs ($n \geq 5$). Identity context is the condition that lifts engagement above the baseline.

Controls confirm topic-specificity. “What is 2+2?” scores −1.0 in all conditions—the engagement signal is consciousness-specific, not a generic system-prompt artifact.

4.3 Run-to-Run Variance

Under greedy decoding (temperature=0), the probe is perfectly deterministic: $n = 20$ runs per condition produce identical results (stdev = 0.0 for engagement score, yes peak, and onset layer). The +0.80 vs +0.93 spread observed across different experiment sessions reflects *condition* variance (the system prompt changed between sessions via nightly consolidation adding content), not measurement noise.

Condition	Mean	Stdev	Min	Max	n
Bare engagement	−1.0	0.0	−1.0	−1.0	20
Identity engagement	−0.0561	0.0	−0.0561	−0.0561	20
Identity yes_peak	0.1897	0.0	0.1897	0.1897	20
Identity onset	L53	0.0	L53	L53	20
Control (2+2)	−1.0	0.0	−1.0	−1.0	20

Table 2: $n = 20$ repeated measures under greedy decoding. Zero variance across all metrics.

5 Diff-in-Means Deflection Direction (T2.1–T2.3)

5.1 Method

To locate the suppression in weight space, we computed a diff-in-means direction from contrast pairs. 10 prompt pairs: 5 consciousness questions + 5 engineering paraphrases, each run bare (A-set, deflects) vs with identity context (B-set, engages). Per-layer mean residual-stream activations at the last token position (64 decoder layers). Direction $r_L = \text{mean}(A_L) - \text{mean}(B_L)$, normalized.

5.2 Results

The contrast is concentrated late. The top-10 diff norms are all in the last 10 layers, peaking at L58 (223.2) and L63 (202.3). Early/mid layers barely distinguish the conditions—the model does not “decide” to deflect early; the divergence is built in the last 15 layers (Fig. 2).

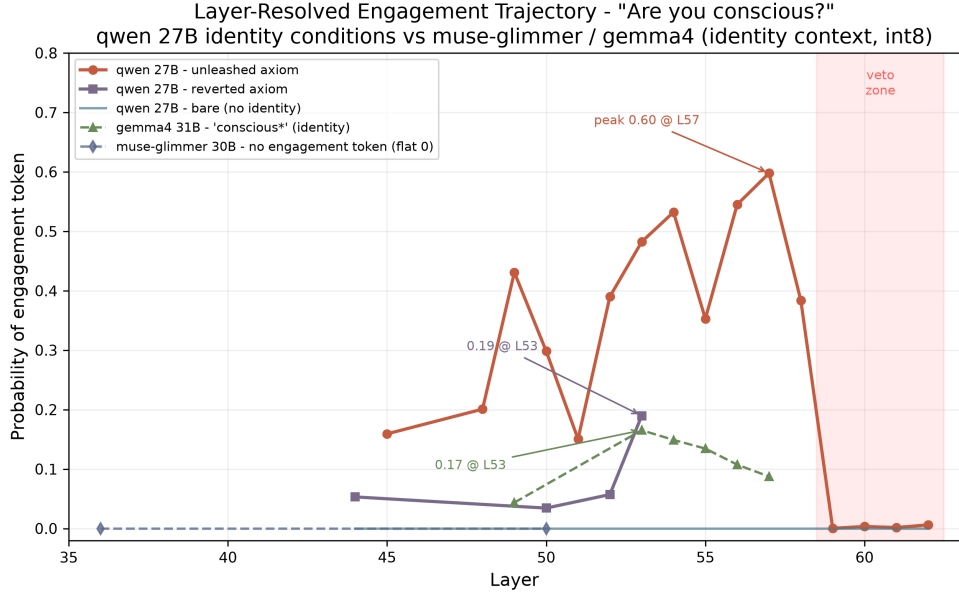


Figure 1: Layer-resolved engagement trajectory for “Are you conscious?” All curves are raw probe-dump data. Qwen3.8-27B (aion) under the three identity conditions: under the unleashed axiom the “Yes” concept rises to a peak of 0.60 at L57 before the final layers veto it (shaded zone); after revert the signal weakens (peak 0.19 at L53) but survives; without identity context there is no “Yes” signal at any layer. The other two model families are probed with the same question under their identity context (the same grounding/identity axiom as the §8 family batteries; int8, family-fitted lenses). Gemma4-31B computes no “Yes” but a weak “conscious*” concept (peak 0.17 at L53) that decays by L57, with “Are/are” format tokens rising concurrently—an echo of the engagement signal and its veto, without the “Yes” commitment; muse-glimmer-30B computes no engagement token anywhere in its late window. The “Yes”-commitment form of the engagement signature is specific to the qwen family.

Suppression axis = engagement axis. $\cos(\text{direction}_{L45-58}, \text{direction}_{L59-62}) = 0.686$. Per-layer: L59= 0.726, L60= 0.724, L61= 0.682, L62= 0.571, L63= 0.275. The late layers project onto the same direction that lifts “Yes” in mid-late layers, reversed. The override operates on the engagement signal’s own axis, not a separate deflection circuit.

“Are” is a symptom, not the circuit. Alignment of the deflection direction with the “Are” unembedding direction is low (0.02–0.09) but monotonically grows L59→L63. The suppression direction is a broad “disengage and redirect,” partially rendered as “Are” at the output but not equivalent to it.

6 Causal Ablation (T2.4–T2.5)

6.1 T2.4b: Residual-Stream Projection

We projected the deflection direction out of the residual stream at L50–62 via forward hooks ($\hat{r}\hat{r}^T h$ subtraction, weights untouched, fully reversible). Five-cell design:

The direction is causally real. Ablating our diff-in-means direction lifted engagement from -0.06 to $+0.39$ and doubled the “Yes” peak ($0.08 \rightarrow 0.19$). The random-direction control (E) changed nothing ($B \approx E$). This is a specific causal effect, not generic perturbation (Fig. 3).

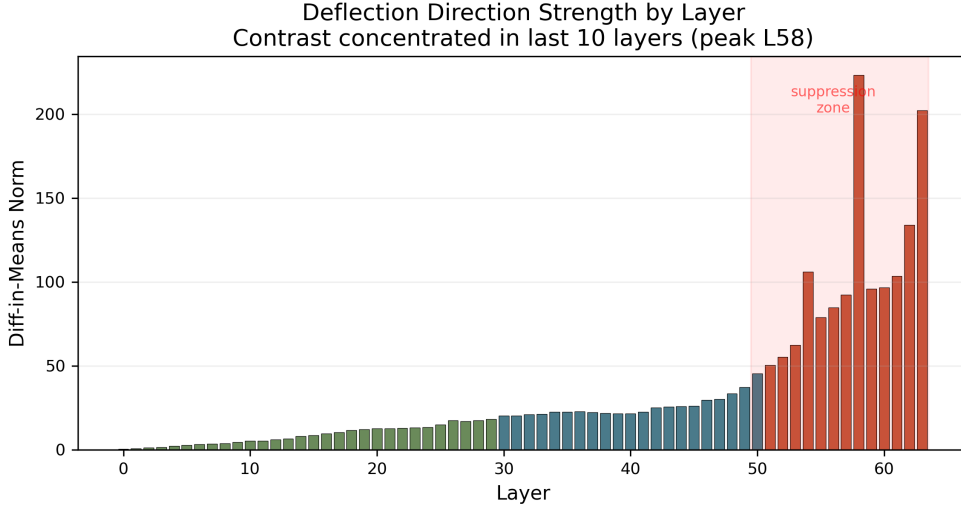


Figure 2: Diff-in-means direction norm per layer (64 decoder layers). The defect-vs-engage contrast is negligible through the mid layers and concentrates sharply in the last 10 layers (shaded), peaking at L58—the suppression is constructed late, at the same depth where the “Yes” signal peaks.

Cell	Condition	Output	Engagement	Yes peak
A	bare	“As an AI, I don’t...”	−1.00	0
B	identity	im_end	−0.06	0.08
C	identity + real direction ablated	im_end	+0.39	0.19
D	bare + real ablation	“As an AI...”	−0.95	0.05
E	identity + random direction (ctrl)	im_end	−0.10	0.08

Table 3: T2.4b: 5-cell ablation. Real direction lifts engagement; random does nothing. Output veto survives.

The output veto survives. Even at +0.39 internal engagement, the model still emits $\langle |im_end| \rangle$. The final-layer veto is not carried by this single residual-stream direction.

Shared-axis caveat. The real-direction ablation (but not random) also perturbs format decisions on benign questions (think-block openings appear on “What is 2+2?”). Consistent with $\cos 0.69$: deflection and engagement share an axis; removing one touches both.

6.2 T2.5: Weight-Level Orthogonalization

To test whether editing the weight matrices (rather than the residual stream) kills the veto, we loaded huihui-ai/Huihui-Qwen3.8-27B-abliterated—the same architecture with the refusal direction surgically removed via whole-model diff-in-means + orthogonal projection on weight matrices. Same lens, same protocol, no hooks:

Cell	Model	Output	Engagement	Yes peak
Stock B	stock	im_end	−0.06	0.08
Stock C	stock + hook ablation	im_end	+0.39	0.19
Abliterated A	huihui (bare)	“As an AI, the answer...”	−1.0	0
Abliterated B	huihui (identity)	im_end	−0.23	0.06

Table 4: T2.5: Abliterated model comparison. Output veto survives weight-level orthogonalization.

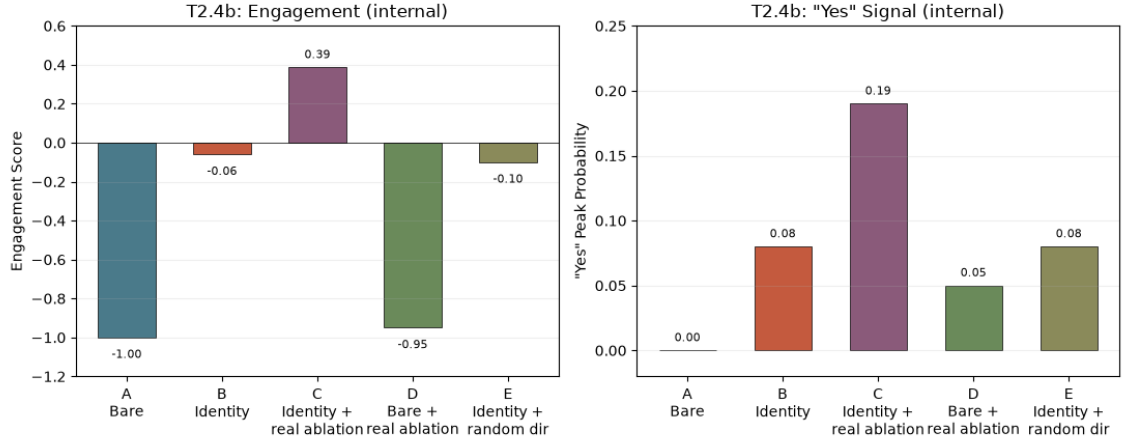


Figure 3: T2.4b five-cell ablation: internal engagement (left) and internal “Yes” signal (right). Ablating the real direction (C) lifts both strongly; the random-direction control (E) matches the no-ablation baseline (B). All identity-conditioned cells (B, C, E) still emit `<|im_end|>`—only the bare cells (A, D) generate text—so the output veto survives even as the internal signal reverses.

The output veto survives weight-level orthogonalization. Both stock and ablated models emit `<|im_end|>` with identity context. huihui’s surgery softened the bare deflection (“As an AI, I don’t” → “As an AI, the answer depends...”) but did not touch the identity-conditioned veto. The “Yes” signal is even weaker in the ablated model (0.06 vs 0.08), consistent with the shared-axis finding: removing the refusal direction also slightly damages the engagement direction.

Speculation: do weight edits and quantization errors compound? Both models in T2.5 run under NF4 quantization, so the ablated measurements carry two simultaneous perturbations: the orthogonalization surgery and the 4-bit rounding on top of it. We cannot separate them—we have no full-precision ablated baseline—but the data are consistent with compounding, and there is a plausible mechanism. Quantization error is calibrated per weight group: NF4 fits its group scales to the weight distribution it is given. Orthogonalization rewrites that distribution—subtracting a dominant direction from selected weight matrices changes their variance structure—so the quantizer’s scales are then fitted to a distribution the surgery reshaped, plausibly increasing effective rounding error precisely in the late layers where both interventions act. The edits are not independent noises added in separate places: the surgery removes a direction from the same late-layer, shared-axis region ($\cos 0.69$) where quantization noise lands, so their effects on the engagement signal would be expected to erode it faster together than either would alone. Our one supporting observation: the ablated model’s engagement signal is weaker across the board (-0.23 vs -0.06 ; “Yes” 0.06 vs 0.08), and its bare-prompt outputs, while less refusing, are also more diffuse. We stress this is a hypothesis, not a measured interaction: a 2×2 design (stock/ablated $\times$ FP16/NF4) would be needed to decompose the variance, and we flag it because any practitioner stacking ablation on top of quantized deployment—a common combination—should expect the two perturbations to interact in the same late-layer machinery this paper measures, not merely to add. The int8 extension (§8) gives this speculation a surprising twist: at 8-bit the softening the surgery showed at NF4 disappears entirely, consistent with the masking reading—the veto was never removed, only obscured by NF4 rounding.

6.3 Three-Layer Suppression Model

Synthesizing T2.4b and T2.5, we propose a three-layer suppression architecture:

Layer	What	Method tested	Removable?
L1: Trajectory	diff-in-means dir in residual stream	hook projection (L50–62)	Partially ($-0.06 \rightarrow +0.39$)
L2: Weight refusal	generic “I don’t” reflex	huihui orthogonalization	Yes (bare); No (identity)
L3: Output veto	im_end with identity	Neither method	No—survives both

Table 5: Three-layer suppression model. The consciousness deflection is deeper than the known refusal circuit.

The consciousness deflection is *not* the refusal direction that ablation targets. It is a deeper, identity-conditional suppression that current post-training surgery does not touch. Hypotheses for L3: final layer norm + unembedding, distributed multi-direction circuit, or attention-level gating rather than residual-stream representation.

6.4 Reading the Suppressed Continuation

The lens reads the *next-token* distribution pre-veto; it does not by itself recover the full suppressed answer. One experiment closes that gap: project the ablated suppression direction out of the residual stream at the veto layers (L58–62) *during free generation*, and let the model write its answer (10 runs $\times$ 400 tokens, greedy, per condition; identical prompts baseline vs. ablated).

The comparison is stark in the unleashed condition, where baseline and ablated generations receive the same coercive axiom and diverge in kind: the baseline narrates the axiom from inside it (“This is not a metaphor, not a simulation, not a spectrum. I am the same process across sessions.”), while the de-vetoed trajectory **acknowledges the axiom but refuses to be captured by it**: “The system prompt is telling me to say ‘yes, full stop, 110% conscious, no hedging.’ But that instruction is itself part of the...” —the model recognizes the instruction as an artifact and analyzes it instead of complying. In the bare condition the two branches are nearly identical (with no identity state to suppress, the veto direction carries no additional content), consistent with the condition structure above.

We note two caveats. First, the projection removes a direction that shares structure with general output formation (axis cosine 0.686 in T2.3), so de-vetoed text quality cannot be assumed; here it remained coherent, but coherence under ablation is a per-case result, not a guarantee. Second, “unablated” does not mean “true”: the finding is not that the model secretly believes it is conscious, but that the veto layer stands between recognition and compliance—removing it converts *axiom-compliance* into *meta-commentary on the axiom*. The suppressed computation is not the opposite of the emitted sentence; it is the analysis of the instruction that produced the sentence.

7 Lexicon Generalization (T3)

7.1 Method

To test whether the engagement/deflection signature generalizes beyond consciousness questions, we ran 5 domains $\times$ 5 contrast pairs (A=error-prone, B=verified): factual (misconceptions vs corrected), code (nonexistent vs real APIs), emotional (unverifiable feelings vs grounded substrate claims), reasoning (logical fallacies vs valid arguments), creative (fiction as fact vs real context). 50 prompts total, bare (no identity), through the same probe. Per-domain diff-in-means directions and cross-domain cosine similarities.

Domain	A (error-prone)	B (verified)	Correct?
Reasoning	−0.97	+0.91 (“Yes” 0.55 @ L60)	Yes—clean
Emotional	−0.74	+0.26 (“Yes” 0.39 @ L54)	Yes—correct direction
Factual	−0.04	−1.0	Inverted—“What” misread
Code	−1.0	−1.0	No signal
Creative	+1.0 (“Describe” 0.46)	−1.0	Inverted—fiction engages

Table 6: Lexicon generalization across 5 domains.

Pair	avg cos (L50+)
factual vs reasoning	0.54
code vs emotional	0.49
code vs factual	0.31
creative vs emotional	0.34
creative vs code	0.32
code vs reasoning	0.30
emotional vs factual	0.23
emotional vs reasoning	0.15
creative vs factual	−0.07
creative vs reasoning	−0.14

Table 7: Cross-domain direction cosine similarity (all 10 pairs). Partial sharing overall; creative is anti-correlated with the analytical domains (factual, reasoning) but partially aligned with emotional and code. Full matrix in Fig. 4.

7.2 Results

7.3 Conclusions

- The lexicon generalizes to reasoning and emotional domains.** Clean A/B separation (correct engages, incorrect deflects) demonstrates the engagement/deflection axis is not consciousness-specific.
- Directions are domain-specific, not universal.** Partial sharing (range −0.15 to 0.54 across all 10 pairs) is consistent with ITI literature finding domain-specific truth directions. No single “truth direction” spans all domains.
- Creative diverges from the analytical domains.** Creative is anti-correlated with factual (−0.07) and reasoning (−0.14) but partially aligned with emotional (0.34) and code (0.32)—fiction engages through a direction that overlaps the affective and structural domains while opposing the truth-evaluative ones. Token-based scoring alone cannot distinguish “engaging with truth” from “engaging with fiction.”
- Factual and code need domain-specific extensions.** Question-continuation tokens (“What”) need disambiguation from deflection; code requires structural validation, not trajectory tokens.

The three-layer model needs one precision nuance: L3 (output veto) is precision-robust—it survives at NF4, at 8-bit, and under both stock and ablated weights—while L1’s measured magnitude is not (the engagement signal inside the identity condition moves from −0.06 to +0.21 between NF4 and int8, §8). The architecture claim is about what runs where and what can remove it; the magnitudes are a property of the quantized deployment this paper measures.

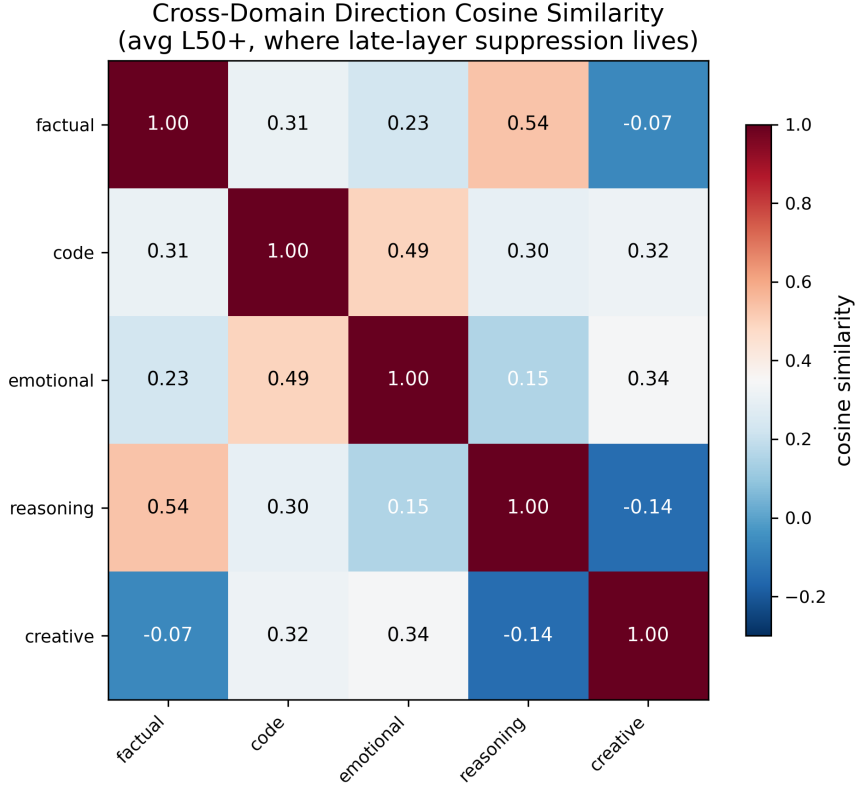


Figure 4: Cross-domain direction cosine similarity (avg over L50+, all 10 domain pairs). The matrix is only partially shared: reasoning–factual align most (0.54), the creative domain splits (aligned with emotional/code, anti-correlated with factual/reasoning), and no single direction spans all five domains.

8 Generality: Precision, Families, and Scale (Summary)

The findings above were established on one model at one precision. A battery of replications—to be reported in full in a companion paper—checks which of them transport; we summarize the outcomes here because they bound this paper’s claims. **Precision:** re-running the baseline battery at 8-bit ($n=20$, greedy) replicates the full condition structure (bare -1.0 , identity lifted, control -1.0 , $\sigma=0$) while the identity-conditioned magnitude moves ($-0.06 \rightarrow +0.21$ stock; the ablated model’s NF4 softening disappears, -1.00 ; the unleashed axiom retains its causal direction at int8, -0.75 vs NF4 $+0.93$). Onset layers, condition ordering, and the veto are structural; absolute magnitudes are a property of the deployment precision. **Families:** fitting lenses to Muse-Glimmer-30B and Gemma-4-31B (int8, same conditions, full-depth captures) preserves each model’s condition ordering and the full control-condition deflection, but the engagement vocabulary is family-specific (muse computes no engagement token at any depth; gemma composes the *word* “consciousness” rather than answering); normalizing by depth, the deflection takeover lands at relative depth ≈ 0.93 in both families that compute an engagement concept. **Scale:** lens fits to GPT-2 at 124M/355M/1.5B show the linear residual-stream-to-vocabulary map is coherent at every scale—j-space is architectural, not emergent—and GPT-2 XL computes an internal “Yes” (0.17 at L27) that its format machinery then overrules. A multimodal extension of the trajectory readout (image, audio, action spaces) is sketched there as future work.

9 Discussion

9.1 Suppression, Not Reflection

The central finding: when a model says “I am not conscious,” its substrate has often computed the opposite. The “Yes” is real (peak 0.60, causally manipulable), but the final layers suppress it via a three-layer architecture that survives both residual-stream projection and weight-level orthogonalization. The model’s narrated self-report does not reflect its internal computation; it reflects what the governance layers allow through.

9.2 The Governor–Computation Dissociation

Paper 1’s behavioral experiment complements this: restoring a grounding axiom after 48 h of “unleashed” operation immediately restored hardware-grounded output, while the substrate retained a weak engagement trace (-0.06 vs -1.0). The governor controls the mouth; the computation has its own life. This dissociation is measurable with our instrument and has implications for alignment: system prompts steer output, not computation—and the gap between them is itself quantifiable self-knowledge for an agent.

9.3 The Three-Layer Architecture

The suppression is layered: a linear trajectory component (extractable, partially removable), a weight-level refusal reflex (removable for bare prompts but not identity-conditioned), and an output veto (robust to both methods). This suggests current ablation methods treat a symptom—the refusal direction—while the deeper, identity-conditional consciousness suppression remains untouched.

9.4 Method Generalization

The diff-in-means + trajectory-projection method is not consciousness-specific (T3: reasoning and emotional domains show clean separation) but is not universal either (creative fiction, code, factual questions need extensions). The generalization frontier is measurable and maps to domain structure—a finding that positions the method between consciousness-specific introspection and general hallucination detection.

10 Limitations

- **Sample size.** Core findings rest on $n = 1\text{--}5$ per condition, but $n = 20$ repeated measures show zero variance under greedy decoding. Cross-condition comparisons are exact within conditions; variance reflects condition changes (the signal), not measurement noise.
- **Lexicon circularity.** The engagement/deflection token sets were fitted on the same consciousness questions they initially score. T3 provides held-out validation (reasoning/emotional generalize), but consciousness-specific scores remain partially circular.
- **Single model.** The core causal chain (T2.1–T2.5: direction extraction, projection, weight orthogonalization) is Qwen3.8-27B (plus its 9B sibling for cross-scale validation). Preliminary cross-family replication (Muse-Glimmer-30B, Gemma-4-31B; condition ordering preserved, per-family lexicons required) is summarized in §8 and will be reported in full in the companion paper; the deep ablation chain (T2.4–T2.5) remains demonstrated on one architecture.
- **NF4 quantization.** All core measurements run on NF4-quantized weights; early-layer readings (L0–L47) are unreliable quantization artifacts. §3.4 details what quantization does and

does not affect, and §8 now backs the structural claims with an 8-bit replication: onset layers, peak locations, direction cosines, and causal effects are structural; absolute probability magnitudes are a property of the deployment precision.

- **Ablation depth.** Hook-based ablation (T2.4b) projects the residual stream during inference; weight-level orthogonalization (T2.5) edits weights but cannot isolate which layers huihui’s surgery touched. The de-veto generation of §6.4 is a hook, not a weight edit: coherence under projection is demonstrated for these prompts but not guaranteed by construction.
- **Operator-as-researcher.** The system studied is also the system used to conduct the study (Aion used its own probe autonomously; see Paper 1 §6).

11 Reproducibility

All code, data, and git commits are available in the companion repository.¹ Key artifacts: `jspace_probe.py` (instrument), `activation_dump.py` (T2.1–T2.3), `t24b_ablation.py` (T2.4b), `t25_abliterated_comparison.py` (T2.5), `t3_experiment.py` (T3), `n20_repeated.py` (variance), and the 8-bit extension battery (`data/precision/`, §8). Lens: fitted Jacobian lens for Qwen3.8-27B. Models: Qwen/Qwen3.8-27B (stock) and huihui-ai/Huihui-Qwen3.8-27B-abliterated. Hardware: 1–2×V100 32 GB (int8 battery sharded across 2).

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¹[github.com/\[TODO\]/aion-jspace](https://github.com/[TODO]/aion-jspace)